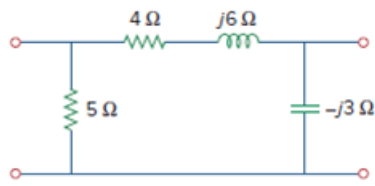


Problem

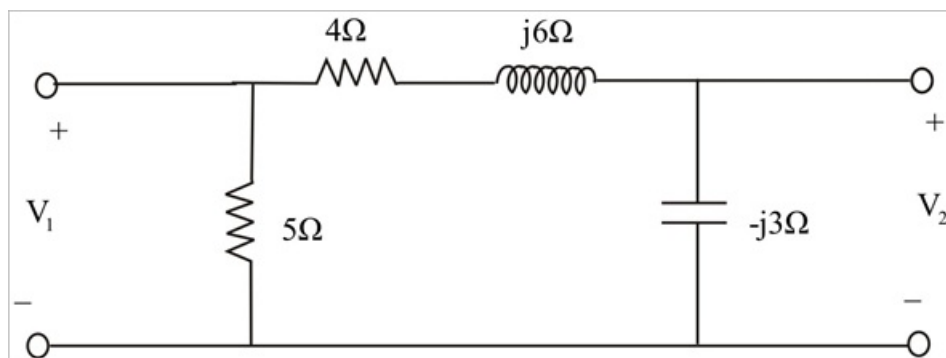
Obtain the h parameters for the two-port of Fig. 19.91

Figure 19.91

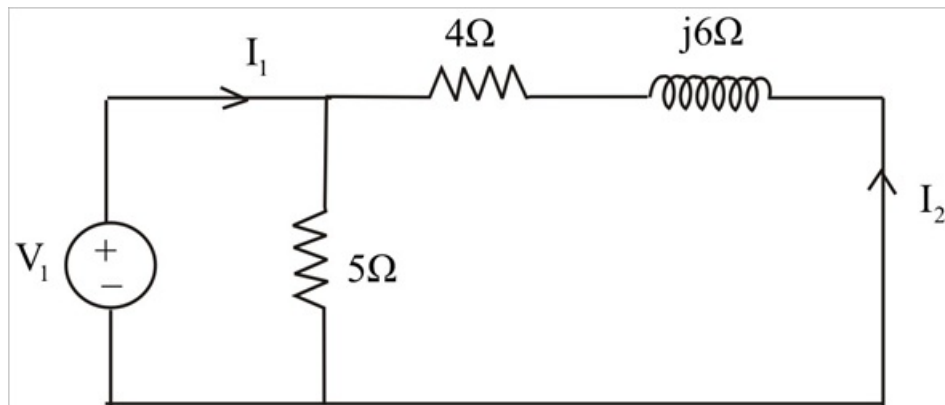


Step-by-step solution

Step 1 of 5



For $V_2 = 0$ we have



Step 2 of 5

$$h_{11} = \left. \frac{V_1}{I_1} \right|_{V_2=0}$$

$$\frac{V_1}{I_1} = \frac{V_1}{\frac{V_1}{(4+6j) \parallel 5}}$$

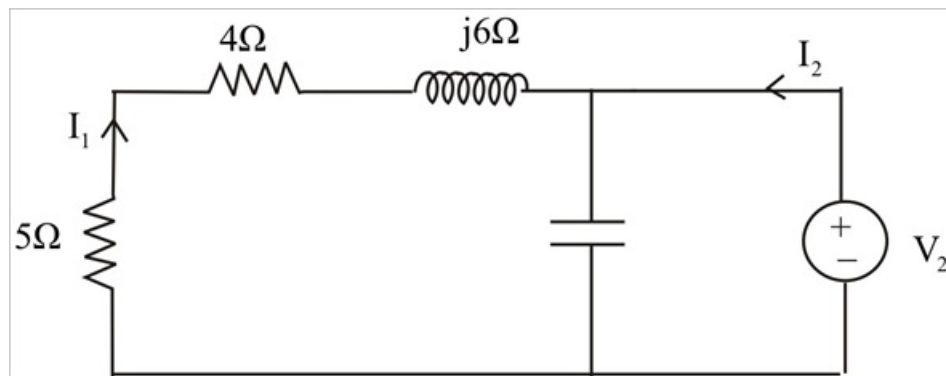
$$\frac{V_1}{I_1} = \frac{(4+6j)5}{9+6j} = \frac{20+30j}{9+6j} = \frac{36.05 \angle 56.3}{10.8 \angle 33.69} = 3.33 \angle 22.61 = 3.07 + 1.28j \, \Omega$$

$$h_{11} = 3.07 + 1.28j \, \Omega$$

$$h_{21} = \left. \frac{I_2}{I_1} \right|_{V_2=0} = \frac{(I_1 \times 5)}{\frac{(5+4+6j)}{I_1}} = \frac{-5}{9+6j} = \frac{-5}{10.8 \angle 33.69^\circ} = -0.385 + 0.2567j$$

Step 3 of 5

For $I_1 = 0$ we have



Step 4 of 5

$$\begin{aligned} \frac{I_2}{V_2} &= \frac{\frac{V_2}{(5+4+6j) \parallel (-j3)}}{V_2} \\ &= \frac{9+3j}{(9+6j)(-3j)} = \frac{9+3j}{-27j+18} \\ \frac{I_2}{V_2} &= \frac{3+j}{6-9j} = \frac{3.16 \angle 18.43}{10.8 \angle -54.630} = 0.2975 \angle 74.73 \\ &= \frac{+j \times 5}{(j+3)(+3j)(9+6j)} = \frac{(j+3)5 \times 3}{3(j+3)3(3+2j)3} \\ &= \frac{1 \times 5 \times 3}{9(3+2j)} \\ \frac{1 \times 5 \times 3}{9 \times 3.6055 \angle 33.69^\circ} &= 145 \times 0.0308 \angle -33.69 \end{aligned}$$

Step 5 of 5

$$h_{12} = \frac{V_1}{V_2} = \frac{I_1 \times 5}{V_2} = \frac{\frac{I_2(-3j)}{3j+9} \times 5}{I_2(9+6j) \parallel (-3j)}$$

$$h = \begin{bmatrix} 3.07 + 1.28j\Omega & -0.385 + 0.2567j \\ 0.3844 - 0.25627j & 0.077 + 0.2821js \end{bmatrix}$$